



COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/17/2026	DATE SUBMITTED:	09/21/2026
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ACTIVITY NO. 8 **[ACTIVITY NAME]**

1. PROCEDURE OUTPUT

Create roles directory for each task using the command `mkdir -p roles/<role>/tasks` from the current directory.

```
qimagalan@workstation: ~/CPE212_Galan/HOA8
qimagalan@workstation: ~/CPE212_Galan/HOA8$ mkdir -p roles/downloader/tasks
qimagalan@workstation: ~/CPE212_Galan/HOA8$ ls -R roles
roles:
creategrp  downloader  initsoftware

roles/creategrp:
tasks

roles/creategrp/tasks:
main.yml

roles/downloader:
tasks

roles/downloader/tasks:

roles/initsoftware:
tasks

roles/initsoftware/tasks:
main.yml
qimagalan@workstation: ~/CPE212_Galan/HOA8$
```



COMPUTER
ENGINEERING

QUEZON CITY

```
qimagalan@workstation: ~/CPE212_Galan/HOA8
qimagalan@workstation: ~/CPE212_Galan/HOA8
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
qimagalan@workstation:~/CPE212_Galan/HOA8$ ansible-playbook install_nagios.yml -i inventory.yml -K
BECOME password:

PLAY [localhost] *****

TASK [Gathering Facts] *****
ok: [localhost]

TASK [initsoftware : Install apache2] *****
ok: [localhost]

TASK [initsoftware : Install PHP] *****
ok: [localhost]

TASK [creategrp : Setup nagios and nagcmd group] *****
ok: [localhost] => (item=nagios)
changed: [localhost] => (item=nagcmd)

TASK [creategrp : Create nagios user] *****
changed: [localhost]

TASK [creategrp : Assign apache2 to nagcmd] *****
changed: [localhost]

PLAY RECAP *****
localhost : ok=6  changed=3  unreachable=0  failed=0  skipped=0  re
scued=0  ignored=0
```

```
qimagalan@workstation: ~/CPE212_Galan/HOA8
qimagalan@workstation: ~/CPE212_Galan/HOA8 x qimagalan@workstation: ~/CPE212_Galan/HOA8 x
ok: [localhost] => (item=nagcmd)

TASK [creategrp : Create nagios user] *****
ok: [localhost]

TASK [creategrp : Assign apache2 to nagcmd] *****
ok: [localhost]

PLAY [localhost] *****

TASK [Gathering Facts] *****
ok: [localhost]

TASK [downloader : Create download destination] *****
changed: [localhost]

TASK [downloader : Download nagios] *****
changed: [localhost]

TASK [downloader : Download nagios plugins] *****
changed: [localhost]

TASK [downloader : Decompress downloaded files] *****
changed: [localhost] => (item=/home/qimagalan/NagiosDownloads/nagios-4.2.1.tar.gz)
changed: [localhost] => (item=/home/qimagalan/NagiosDownloads/nagios-plugins-2.1.3.tar.gz)

PLAY RECAP *****
localhost : ok=11 changed=4 unreachable=0 failed=0 skipped=0 re
scued=0 ignored=0

qimagalan@workstation:~/NagiosDownloads$ ls
nagios-4.2.1 nagios-4.2.1.tar.gz nagios-plugins-2.1.3.tar.gz nagios-plugins-release-2.1.3
```

Create the installer for software package dependencies under the role initsoftware

```
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~ x
GNU nano 7.2 roles/initsoftware/tasks/main.yml
---
- name: Install nagios dependencies (Ubuntu)
  apt:
    name:
      - apache2
      - libapache2-mod-php
      - libgd-dev
      - build-essential
      - openssl
      - libssl-dev
      - python3-passlib
    state: present
  when: ansible_distribution == "Ubuntu"

- name: Install nagios dependencies (CentOS)
  dnf:
    name:
      - httpd
      - php
      - gcc
      - glibc-common
      - gd
      - gd-devel
      - openssl
      - openssl-devel
      - python3-passlib
      - chkconfig
    state: present
  when: ansible_distribution == "CentOS"
```

Create the playbook that would execute group creations and assignment in the creategrp role

```
qimagalan-workstation@workstation: ~/CPE212_Galan/...
GNU nano 7.2 creategrp/tasks/main.yml
--
- name: Setup nagios and nagcmd group
  ansible.builtin.group:
    name: "{{ item }}"
    state: present
    system: true
  loop:
    - nagios
    - nagcmd

- name: Create nagios user
  ansible.builtin.user:
    name: nagios
    home: /home/nagios
    create_home: true
    shell: /bin/bash
    system: true
    state: present
    groups:
      - nagios
      - nagcmd
    append: true

- name: Assign apache2 to nagcmd
  ansible.builtin.user:
    name: www-data
    groups:
      - nagcmd
    append: true

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

Create the downloader playbook under the downloader role

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA8
GNU nano 7.2 roles/downloader/tasks/main.yml

- name: Create download destination
  become: false
  ansible.builtin.file:
    path: "{{ ansible_env.HOME }}/NagiosDownloads"
    state: directory

- name: Download nagios
  become: false
  ansible.builtin.get_url:
    url: "https://github.com/NagiosEnterprises/nagioscore/archive/refs/tags/nagios-4.5.12.tar.gz"
    dest: "{{ ansible_env.HOME }}/NagiosDownloads/nagios-4.5.12.tar.gz"

- name: Download nagios plugins
  become: false
  ansible.builtin.get_url:
    url: "https://github.com/nagios-plugins/nagios-plugins/releases/download/release-2.4.8/nagios-plugins-2.4.8.tar.gz"
    dest: "{{ ansible_env.HOME }}/NagiosDownloads/nagios-plugins-2.4.8.tar.gz"

- name: Decompress downloaded files
  become: false
  ansible.builtin.unarchive:
    src: "{{ item }}"
    dest: "{{ ansible_env.HOME }}/NagiosDownloads"
    remote_src: true
  loop:
    - "{{ ansible_env.HOME }}/NagiosDownloads/nagios-4.5.12.tar.gz"
    - "{{ ansible_env.HOME }}/NagiosDownloads/nagios-plugins-2.4.8.tar.gz"
```

Test whether installation procedures works by creating the main playbook to run the roles:

```
qimagalan-workstation@workstation: ... x qimagalan-workstation@workstation: ... x
GNU nano 7.2 install_nagios.yml
--
- hosts:
  - localhost
  - file_servers
  become: true

pre_tasks:
  - name: Update repository list (Ubuntu)
    apt:
      update_cache: true
    when: ansible_distribution == "Ubuntu"

  - name: Update repository list (CentOS)
    dnf:
      update_cache: true
    when: ansible_distribution == "CentOS"

- hosts:
  - localhost
  - file_servers
  become: true
  roles:
    - initsoftware
    - creategrp

- hosts:
  - localhost
  - file_servers
  roles:
    - downloader
#   - compiler
```

Then run using:
ansible-playbook install_nagios.yml -K



```
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$ ansible-playbook install_nagios.yml -i inventory
yml -K
BECOME password:

PLAY [localhost,file_servers] *****

TASK [Gathering Facts] *****
ok: [localhost]
ok: [192.168.56.114]

TASK [initsoftware : Install apache2 (Ubuntu)] *****
skipping: [192.168.56.114]
ok: [localhost]

TASK [initsoftware : Install apache2 (CentOS)] *****
skipping: [localhost]
ok: [192.168.56.114]

TASK [initsoftware : Install PHP (CentOS)] *****
ok: [192.168.56.114]

TASK [creategrp : Setup nagios and nagcmd group] *****
changed: [192.168.56.114] => (item=nagios)
changed: [192.168.56.114] => (item=nagcmd)

TASK [creategrp : Create nagios user] *****
changed: [192.168.56.114]

TASK [creategrp : Assign apache2 to nagcmd] *****
changed: [192.168.56.114]

PLAY [localhost,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]

TASK [downloader : Create download destination] *****
changed: [192.168.56.114]
```

```
TASK [downloader : Create download destination] *****
changed: [localhost]
ok: [192.168.56.114]

TASK [downloader : Download nagios] *****
changed: [localhost]
ok: [192.168.56.114]

TASK [downloader : Download nagios plugins] *****
ok: [192.168.56.114]
changed: [localhost]

TASK [downloader : Decompress downloaded files] *****
changed: [localhost] => (item=/home/qimagalan-workstation/NagiosDownloads/nagios-4.2.1.tar.gz)
changed: [localhost] => (item=/home/qimagalan-workstation/NagiosDownloads/nagios-plugins-2.1.3.tar.gz)
changed: [192.168.56.114] => (item=/home/qimagalan/NagiosDownloads/nagios-4.2.1.tar.gz)
changed: [192.168.56.114] => (item=/home/qimagalan/NagiosDownloads/nagios-plugins-2.1.3.tar.gz)

PLAY RECAP *****
192.168.56.114      : ok=13   changed=1   unreachable=0    failed=0    skipped=3    rescued=0
ignored=0
localhost          : ok=13   changed=9   unreachable=0    failed=0    skipped=3    rescued=0
ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$
```

Expected Results:

```
qimagalan-workstation@workstation: ... x qimagalan-workstation@workstation: ... x qimagalan-workstation@workstation: ... x
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$ ls ~/NagiosDownloads
nagios-4.2.1 nagios-4.2.1.tar.gz nagios-plugins-2.1.3.tar.gz nagios-plugins-release-2.1.3
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$

qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$ ssh qimagalan@Galan_CentOS
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Sep 21 11:09:24 2026 from 192.168.56.1
[qimagalan@localhost ~]$ ls ~/NagiosDownloads
nagios-4.2.1 nagios-4.2.1.tar.gz nagios-plugins-2.1.3.tar.gz nagios-plugins-release-2.1.3
[qimagalan@localhost ~]$

qimagalan-workstation@workstation: ~/CPE212_Galan/HOA8
qimagalan-workstation@workstation: ... x qimagalan-workstation@workstation: ... x qimagalan@lo
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$ cat /etc/passwd | grep nagios
nagios:x:997:100:~/home/nagios:/bin/bash
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$ cat /etc/group | grep nagcmd
nagcmd:x:983:nagios,www-data
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$
```

```
qimagalan-workstation@workstation: ... x qimagalan-workstation@workstation: ... x
[qimagalan@localhost ~]$ cat /etc/passwd | grep nagios
nagios:x:983:100::/home/nagios:/bin/bash
[qimagalan@localhost ~]$ cat /etc/group | grep nagcmd
nagcmd:x:982:nagios,www-data
[qimagalan@localhost ~]$
```

Create the playbook that would compile the nagios source

```
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~ x qimagalan-workstation@work... x
GNU nano 7.2 roles/compiler/tasks/main.yml
---
- name: Retain home directory before privilege escalation
  become: false
  ansible.builtin.set_fact: #Prevents the directory from defaulting to root during become: true
    normal_home: "{{ ansible_env.HOME }}"

- name: Run nagios configure script
  ansible.builtin.command:
    cmd: ./configure --with-command-group=nagcmd
    chdir: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12"
    creates: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12/Makefile" #ensures config only runs once

- name: Compile Nagios Core
  ansible.builtin.command:
    cmd: make all
    chdir: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12"
    creates: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12/base/nagios"

- name: Compile Nagios binaries
  become: true
  ansible.builtin.command:
    cmd: "make {{ item }}"
    chdir: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12"
  loop:
    - install
    - install-init
    - install-config
    - install-commandmode
```

Test out the compiler playbook

In install_nagios.yml add the last block with compiler role

```
- hosts:
  - localhost
  - file_servers
become: true
roles:
  - initsoftware
  - creategrp

- hosts:
  - localhost
  - file_servers
roles:
  - downloader

- hosts:
  - localhost
  - file_servers
become: true
roles:
  - compiler
```

```
PLAY [localhost,file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]
ok: [localhost]

TASK [compiler : Retain home directory before privilege escalation] *****
ok: [localhost]
ok: [192.168.56.114]

TASK [compiler : Run nagios configure script] *****
changed: [localhost]
changed: [192.168.56.114]

TASK [compiler : Compile Nagios Core] *****
changed: [192.168.56.114]
changed: [localhost]

TASK [compiler : Compile Nagios binaries] *****
changed: [localhost] => (item=install)
changed: [localhost] => (item=install-init)
changed: [localhost] => (item=install-config)
changed: [localhost] => (item=install-commandmode)
changed: [192.168.56.114] => (item=install)
changed: [192.168.56.114] => (item=install-init)
changed: [192.168.56.114] => (item=install-config)
changed: [192.168.56.114] => (item=install-commandmode)

PLAY RECAP *****
192.168.56.114      : ok=17   changed=4    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
localhost          : ok=17   changed=8    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$
```

Configure the email to receive alerts from nagios with a simple `sudo nano /usr/local/nagios/etc/objects/contacts.cfg`

```
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~ x qimagalan-workstation@work... x
GNU nano 7.2 /usr/local/nagios/etc/objects/contacts.cfg

# Just one contact defined by default - the Nagios admin (that's you)
# This contact definition inherits a lot of default values from the
# 'generic-contact' template which is defined elsewhere.

define contact {

    contact_name      nagiosadmin          ; Short name of user
    use               generic-contact      ; Inherit default values from generic-contact template (defined
    alias            Nagios Admin          ; Full name of user
    email            qimagalan@tip.edu.ph ; <***** CHANGE THIS TO YOUR EMAIL ADDRESS *****
}

#####
#
# CONTACT GROUPS
#
#####

# We only have one contact in this simple configuration file, so there is
# no need to create more than one contact group.

define contactgroup {

    contactgroup_name  admins
    alias              Nagios Administrators
    members            nagiosadmin
}

[ Wrote 51 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

In addition to nagios core compiler, also add on the bottom of the tasks file, a compiler for the plugins under the role compiler

```
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~ x qimagalan-workstation@work... x
GNU nano 7.2 roles/compiler/tasks/main.yml *

- name: Compile Nagios binaries
  become: true
  ansible.builtin.command:
    cmd: "make {{ item }}"
    chdir: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12"
  loop:
    - install
    - install-init
    - install-config
    - install-commandmode

- name: Configure Plugins
  ansible.builtin.command:
    cmd: ./configure --with-nagios-user=nagios --with-nagios-group=nagios
    chdir: "{{ normal_home }}/NagiosDownloads/nagios-plugins-2.4.8"
    creates: "{{ normal_home }}/NagiosDownloads/nagios-plugins-2.4.8/Makefile"

- name: Compile Plugins
  ansible.builtin.command:
    cmd: make
    chdir: "{{ normal_home }}/NagiosDownloads/nagios-plugins-2.4.8"
    creates: "{{ normal_home }}/NagiosDownloads/nagios-plugins-2.4.8/Installed"

- name: Install Plugins
  become: true
  ansible.builtin.command:
    cmd: make install
    chdir: "{{ normal_home }}/NagiosDownloads/nagios-plugins-2.4.8"
    creates: "{{ normal_home }}/NagiosDownloads/nagios-plugins-2.4.8/Installed"
```

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA8
ok: [192.168.56.114]

TASK [compiler : Compile Nagios Core] *****
ok: [localhost]
ok: [192.168.56.114]

TASK [compiler : Compile Nagios binaries] *****
changed: [localhost] => (item=install)
changed: [localhost] => (item=install-init)
changed: [localhost] => (item=install-config)
changed: [localhost] => (item=install-commandmode)
changed: [192.168.56.114] => (item=install)
changed: [192.168.56.114] => (item=install-init)
changed: [192.168.56.114] => (item=install-config)
changed: [192.168.56.114] => (item=install-commandmode)

TASK [compiler : Configure Plugins] *****
changed: [localhost]
changed: [192.168.56.114]

TASK [compiler : Compile Plugins] *****
changed: [localhost]
changed: [192.168.56.114]

TASK [compiler : Install Plugins] *****
changed: [localhost]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.114      : ok=20   changed=5    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
localhost          : ok=20   changed=5    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$
```

Configure the user login for the web UI of Nagios

```
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~
GNU nano 7.2 roles/acctconfigure/tasks/main.yml
---
- name: Retain home directory before privilege escalation
  ansible.builtin.set_fact:
    normal_home: "{{ ansible_env.HOME }}"

- name: Install the Web Interface for Nagios
  become: true
  ansible.builtin.command:
    cmd: "make install-webconf"
    chdir: "{{ normal_home }}/NagiosDownloads/nagioscore-nagios-4.5.12"
    creates: "/etc/apache2/sites-enabled/nagios.conf"

- name: Create nagios http admin for web interface
  become: true
  community.general.htpasswd:
    path: /usr/local/nagios/etc/htpasswd.users
    name: nagiosadmin
    password: "nagiosadmin"
    owner: root
    group: www-data
    mode: "0640"
```

Test out the web interface user creation

```
TASK [acctconfigure : Retain home directory before privilege escalation] *****
ok: [localhost]
ok: [192.168.56.114]

TASK [acctconfigure : Install the Web Interface for Nagios] *****
ok: [localhost]
changed: [192.168.56.114]

TASK [acctconfigure : Create nagios http admin for web interface] *****
changed: [localhost]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.114      : ok=23  changed=6  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0
localhost          : ok=23  changed=5  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan/H0A8$
```

Restart Apache Service and Start Nagios

```
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~ x
GNU nano 7.2 roles/starter/tasks/main.yml
--
- name: Restart Apache service after account configuration (Ubuntu)
  ansible.builtin.service:
    name: apache2
    state: restarted
  when: ansible_distribution == "Ubuntu"

- name: Restart Apache service affter account configuration (CentOS)
  ansible.builtin.service:
    name: httpd
    state: restarted
  when: ansible_distribution == "CentOS"

- name: Start and Enable Nagios Core
  ansible.builtin.service:
    name: nagios
    state: started
    enabled: true

TASK [Gathering Facts] *****
ok: [localhost]
ok: [192.168.56.114]

TASK [starter : Restart Apache service after account configuration (Ubuntu)] *****
skipping: [192.168.56.114]
changed: [localhost]

TASK [starter : Restart Apache service affter account configuration (CentOS)] *****
skipping: [localhost]
changed: [192.168.56.114]

TASK [starter : Start and Enable Nagios Core] *****
ok: [localhost]
changed: [192.168.56.114]

PLAY RECAP *****
192.168.56.114      : ok=26  changed=8  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0
localhost          : ok=26  changed=5  unreachable=0  failed=0  skipped=3  rescued=0  ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan/H0A8$
```

Final main installer yml (install_nagios.yml)

GNU nano 7.2

install_nagios.yml

```
--
- hosts:
  - localhost
  - file_servers
  become: true

  pre_tasks:
    - name: Update repository list (Ubuntu)
      apt:
        update_cache: true
      when: ansible_distribution == "Ubuntu"

    - name: Update repository list (CentOS)
      dnf:
        update_cache: true
      when: ansible_distribution == "CentOS"

- hosts:
  - localhost
  - file_servers
  become: true
  roles:
    - initsoftware
    - creategrp

- hosts:
  - localhost
  - file_servers
  roles:
    - downloader
```

```
- hosts:
  - localhost
  - file_servers
  roles:
    - compiler
    - acctconfigure

- hosts:
  - localhost
  - file_servers
  become: true
  roles:
    - starter
```

Verify by installation by going through <http://localhost/nagios/>

Galan_Workstation (Before Nagios 2) [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

Sep 21 15:01

Nagios: localhost

http://localhost/nagios/

VPN

Nagios

General

Home

Documentation

Current Status

Tactical Overview

Map

Hosts

Services

Host Groups

Summary

Grid

Service Groups

Summary

Grid

Problems

Services (Unhandled)

Hosts (Unhandled)

Network Outages

Quick Search:

Reports

Availability

Trends

Alerts

History

Summary

Histogram

Notifications

Event Log

System

Comments

Downtime

Process Info

Version 4.5.12

March 25, 2026

Not running

Check for updates

Dashboard

Server Statistics

Status Grid

Current Status

Administrative Tools

Meet Nagios Core Services Platform

The next generation of Open Source powered monitoring with advanced dashboards, monitoring

Page Tour



COMPUTER
ENGINEERING

QUEZON CITY

```
Sep 21 15:11
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA8
qimagalan-workstation@work... x qimagalan-workstation@work... x qimagalan@localhost:~ x qimagalan-workstation@work... x
qimagalan-workstation@workstation:~/CPE212_Galan/HOA8$ sudo systemctl status nagios
Warning: The unit file, source configuration file or drop-ins of nagios.service changed on disk. Run 'systemctl daemon-reload' to reload units.
● nagios.service - Nagios Core 4.5.12
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-09-21 14:58:35 PST; 13min ago
     Docs: https://www.nagios.org/documentation
   Process: 80521 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg (code=exited, status=0)
   Process: 80524 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg (code=exited, status=0)
   Main PID: 80529 (nagios)
    Tasks: 8 (limit: 3459)
   Memory: 7.4M (peak: 9.9M swap: 268.0K swap peak: 268.0K)
      CPU: 1.336s
   CGroup: /system.slice/nagios.service
           └─80529 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             └─80531 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
               └─80532 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                 └─80533 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                   └─80534 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                     └─80535 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                       └─80536 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                         └─80546 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Sep 21 14:58:35 workstation nagios[80529]: wproc: Registry request: name=Core Worker 80532;pid=80532
Sep 21 14:58:35 workstation nagios[80529]: wproc: Registry request: name=Core Worker 80533;pid=80533
Sep 21 14:58:35 workstation nagios[80529]: wproc: Registry request: name=Core Worker 80531;pid=80531
Sep 21 14:58:35 workstation nagios[80529]: wproc: Registry request: name=Core Worker 80535;pid=80535
Sep 21 14:58:35 workstation nagios[80529]: wproc: Registry request: name=Core Worker 80536;pid=80536
Sep 21 14:58:39 workstation nagios[80529]: Successfully launched command file worker with pid 80546
Sep 21 15:02:20 workstation nagios[80529]: SERVICE ALERT: localhost;SSH;CRITICAL;SOFT;1;connect to address 127.0.0.1
Sep 21 15:03:20 workstation nagios[80529]: SERVICE ALERT: localhost;SSH;CRITICAL;SOFT;2;connect to address 127.0.0.1
Sep 21 15:04:21 workstation nagios[80529]: SERVICE ALERT: localhost;SSH;CRITICAL;SOFT;3;connect to address 127.0.0.1
Sep 21 15:05:21 workstation nagios[80529]: SERVICE ALERT: localhost;SSH;CRITICAL;HARD;4;connect to address 127.0.0.1
lines 1-31/31 (END)
```

```
[qimagalan@localhost ~]$ sudo systemctl nagios
[sudo] password for qimagalan:
^[A^Csudo: a password is required
[qimagalan@localhost ~]$ sudo systemctl status nagios
[sudo] password for qimagalan:
● nagios.service - Nagios Core 4.5.12
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-21 15:10:16 PST; 2min 8s ago
     Docs: https://www.nagios.org/documentation
  Process: 34958 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg (code=exited, status=0)
  Process: 34961 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg (code=exited, status=0)
 Main PID: 34962 (nagios)
    Tasks: 6 (limit: 10500)
   Memory: 10.1M (peak: 12.2M)
      CPU: 486ms
   CGroup: /system.slice/nagios.service
           └─34962 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             └─34963 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
               └─34964 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                 └─34965 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                   └─34966 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                     └─34995 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

Sep 21 15:10:16 localhost.localdomain nagios[34962]: qh: core query handler registered
Sep 21 15:10:16 localhost.localdomain nagios[34962]: qh: echo service query handler registered
Sep 21 15:10:16 localhost.localdomain nagios[34962]: qh: help for the query handler registered
Sep 21 15:10:16 localhost.localdomain nagios[34962]: wproc: Successfully registered manager as @wproc with query handler
Sep 21 15:10:16 localhost.localdomain nagios[34962]: wproc: Registry request: name=Core Worker 34963;pid=34963
Sep 21 15:10:16 localhost.localdomain nagios[34962]: wproc: Registry request: name=Core Worker 34964;pid=34964
Sep 21 15:10:16 localhost.localdomain nagios[34962]: wproc: Registry request: name=Core Worker 34965;pid=34965
Sep 21 15:10:16 localhost.localdomain nagios[34962]: wproc: Registry request: name=Core Worker 34966;pid=34966
Sep 21 15:10:19 localhost.localdomain nagios[34962]: Successfully launched command file worker with pid 34995
Sep 21 15:12:08 localhost.localdomain nagios[34962]: SERVICE ALERT: localhost;HTTP;WARNING;SOFT;1;HTTP WARNING: HTTP
lines 1-28/28 (END)
```

Create a github installer repository

```
qimagalan-workstation@workstation:~$ git clone git@github.com:IrickMarvinGalan/Galan_NagiosInstaller.git
Cloning into 'Galan_NagiosInstaller'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
qimagalan-workstation@workstation:~$

qimagalan-workstation@workstation:~/Galan_NagiosInstaller$ cp -r ~/CPE212_Galan/HOA8/* ~/Galan_NagiosInstaller

qimagalan-workstation@workstation:~/Galan_NagiosInstaller$ ls
install_nagios.yml  inventory.yml  roles
qimagalan-workstation@workstation:~/Galan_NagiosInstaller$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        install_nagios.yml
        inventory.yml
        roles/
```

```
qimagalan-workstation@workstation:~/Galan_NagiosInstaller$ git add .
qimagalan-workstation@workstation:~/Galan_NagiosInstaller$ git commit -m "Nagios Installer for Ubuntu and CentOS"
[main 824bbbb] Nagios Installer for Ubuntu and CentOS
9 files changed, 230 insertions(+), 2 deletions(-)
delete mode 100644 README.md
create mode 100644 install_nagios.yml
create mode 100644 inventory.yml
create mode 100644 roles/acctconfigure/tasks/main.yml
create mode 100644 roles/compiler/tasks/main.yml
create mode 100644 roles/creategrp/tasks/main.yml
create mode 100644 roles/downloader/tasks/main.yml
create mode 100644 roles/initsoftware/tasks/main.yml
create mode 100644 roles/starter/tasks/main.yml
qimagalan-workstation@workstation:~/Galan_NagiosInstaller$ git push origin main
Enumerating objects: 24, done.
Counting objects: 100% (24/24), done.
Delta compression using up to 4 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (23/23), 3.15 KiB | 645.00 KiB/s, done.
Total 23 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:IrickMarvinGalan/Galan_NagiosInstaller.git
8cbbb12..824bbbb main -> main
qimagalan-workstation@workstation:~/Galan_NagiosInstaller$
```

Test Out the github installer on a VM without nagios

Download the github repository

wget https://github.com/IrickMarvinGalan/Galan_NagiosInstaller/archive/refs/heads/main.tar.gz

```
qimagalan-workstation@workstation:~$ wget https://github.com/IrickMarvinGalan/Galan_NagiosInstaller/archive/refs/heads/main.tar.gz
--2026-09-21 10:53:47-- https://github.com/IrickMarvinGalan/Galan_NagiosInstaller/archive/refs/heads/main.tar.gz
Resolving github.com (github.com)... 20.205.243.166
Connecting to github.com (github.com)|20.205.243.166|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/IrickMarvinGalan/Galan_NagiosInstaller/tar.gz/refs/heads/main [following]
--2026-09-21 10:53:47-- https://codeload.github.com/IrickMarvinGalan/Galan_NagiosInstaller/tar.gz/refs/heads/main
Resolving codeload.github.com (codeload.github.com)... 20.205.243.165
Connecting to codeload.github.com (codeload.github.com)|20.205.243.165|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [application/x-gzip]
Saving to: 'main.tar.gz'

main.tar.gz          [ <=>          ]  2.10K  --.-KB/s    in 0s

2026-09-21 10:53:48 (4.25 MB/s) - 'main.tar.gz' saved [2152]
```

```
qimagalan-workstation@workstation: ~/Downloads

qimagalan-workstation@workstation:~/Downloads$ ls
main.tar.gz  terraform_0.12.28_linux_amd64.zip
qimagalan-workstation@workstation:~/Downloads$
```

Extract the nagios installer

tar xzf main.tar.gz

```
qimagalan-workstation@workstation:~/Downloads$ tar -xzf main.tar.gz
Galan_NagiosInstaller-main/
Galan_NagiosInstaller-main/install_nagios.yml
tar: Galan_NagiosInstaller-main/install_nagios.yml: time stamp 2026-09-21 15:21:
39 is 15957.447769437 s in the future
Galan_NagiosInstaller-main/inventory.yml
tar: Galan_NagiosInstaller-main/inventory.yml: time stamp 2026-09-21 15:21:39 is
15957.447449725 s in the future
Galan_NagiosInstaller-main/roles/
Galan_NagiosInstaller-main/roles/acctconfigure/
Galan_NagiosInstaller-main/roles/acctconfigure/tasks/
Galan_NagiosInstaller-main/roles/acctconfigure/tasks/main.yml
tar: Galan_NagiosInstaller-main/roles/acctconfigure/tasks/main.yml: time stamp 2
026-09-21 15:21:39 is 15957.446625021 s in the future
Galan_NagiosInstaller-main/roles/compiler/
tar: Galan_NagiosInstaller-main/roles/acctconfigure/tasks: time stamp 2026-09-21
15:21:39 is 15957.446580648 s in the future
tar: Galan_NagiosInstaller-main/roles/acctconfigure: time stamp 2026-09-21 15:21
:39 is 15957.446556873 s in the future
Galan_NagiosInstaller-main/roles/compiler/tasks/
Galan_NagiosInstaller-main/roles/compiler/tasks/main.yml
tar: Galan_NagiosInstaller-main/roles/compiler/tasks/main.yml: time stamp 2026-0
9-21 15:21:39 is 15957.44547746 s in the future
Galan_NagiosInstaller-main/roles/creategrp/
tar: Galan_NagiosInstaller-main/roles/compiler/tasks: time stamp 2026-09-21 15:2
1:39 is 15957.444870888 s in the future
tar: Galan_NagiosInstaller-main/roles/compiler: time stamp 2026-09-21 15:21:39 i
s 15957.444785858 s in the future
Galan_NagiosInstaller-main/roles/creategrp/tasks/
Galan_NagiosInstaller-main/roles/creategrp/tasks/main.yml
tar: Galan_NagiosInstaller-main/roles/creategrp/tasks/main.yml: time stamp 2026-
09-21 15:21:39 is 15957.444490461 s in the future
Galan_NagiosInstaller-main/roles/downloader/
tar: Galan_NagiosInstaller-main/roles/creategrp/tasks: time stamp 2026-09-21 15:
```

Notice how roles started to get unpacked implying successful installation of the required ansible-playbooks

Go inside Galan_NagiosIntaller-main directory then do:

```
qimagalan-workstation@workstation:~$ cd Galan_NagiosInstaller-main
qimagalan-workstation@workstation:~/Galan_NagiosInstaller-main$ ls
install_nagios.yml  inventory.yml  roles
```

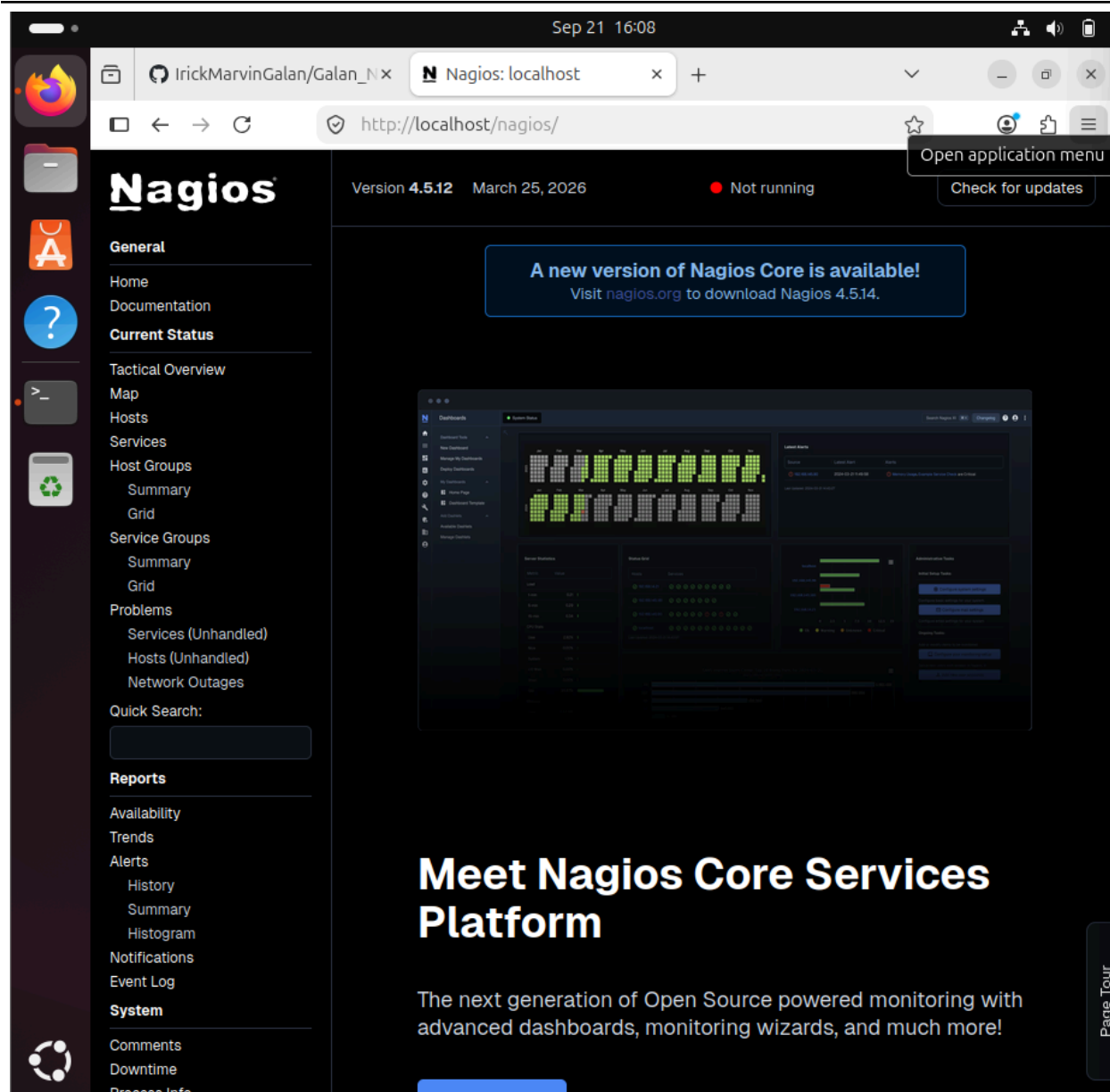
ansible-playbook install_nagios.yml -K -i inventory.yml

```
PLAY RECAP *****
192.168.56.114      : ok=26  changed=18  unreachable=0  failed=0  s
kipped=3    rescued=0    ignored=0
localhost    : ok=1   changed=0  unreachable=0  failed=1  s
kipped=0    rescued=0    ignored=0
```

```
PLAY RECAP *****
192.168.56.114      : ok=18  changed=2  unreachable=1  failed=0  s
kipped=2    rescued=0    ignored=0
localhost    : ok=26  changed=19  unreachable=0  failed=0  s
kipped=3    rescued=0    ignored=0
```

```
qimagalan-workstation@workstation:~/Galan_NagiosInstaller-main$
```

The ubuntu had a temporary update bug in the first run, and CentOS encountered a host unreachable bug in the second run. Overall, after two runs both Ubuntu and CentOS now has the nagios.



Nagios Repository Github: https://github.com/IrickMarvinGalan/Galan_NagiosInstaller

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

The benefit that monitoring tools is that they provide an overview of the machine/server states in a single machine. Doing ssh and checkups on every single machine would not be necessary. Thus, this is extremely beneficial for applications that require strict reliability with uptime, because Monitoring Tools like nagios, provide system administrators a summary of when and where a server error occurs. It is able to notify administrators as well through email such as how it is part of the initial nagios configuration.

Conclusions/Learnings

In this activity I learned a lot of things by compiling source code into usable web interface apps like Nagios. I learned how to create a system user through `ansible.builtin.user`. I also learned that for shell commands, we can use the `ansible.builtin.command`. The important part here is that we use the `chdir` variable in order to locate the installed nagioscore files and nagiosplugin files, because we need to be inside their locations to do the make install commands. I also learned that we should control and be careful the usage of `become: true` because it messes with the `ansible_env.HOME` directory, as it changes it to root's home, and therefore we need to save the unescalated home path first so we could use it while in root. One thing I will improve in this installer is the use of vars, I have not really applied vars to simplify the path directories in one place which takes tedious effort just to modify path when a new update of nagios comes through, making it a bit unscalable in the long run.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.